

Calculation of phonon spectrum within NEMO-3D code

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Abstract

This note contains a brief description of the model, input file and refers to some examples of calculation of phonon spectrum.

1 Model

The current version of the code is dealing with two-parameter Keating model for calculation of the phonon spectrum. It may be applicable to covalent semiconductors (C, Si, Ge, etc.) with 10% accuracy in the whole Brillouin zone, except the lack in reproducing flattening of transversal acoustical dispersion near the Brillouin zone boundary. The model does not include Coulomb interaction, which leads to splitting of optical phonon states in the center of the Brillouin zone in bulk A_3B_5 semiconductors. Thus phonon dispersion, computed for A_3B_5 material systems, may serve as a preliminary estimation only.

1.1 Valence-force-field Keating model

The simplest way to deal with crystall lattice on atomic level is to model it by a set of small balls of certain mass, corresponding to different atoms, connected with springs that represent atomic bonds.

Two-parameter Keating model consider only nearest-neighbour atoms interactions. The original Keating [?] formula that was rewritten later by Martin [?] for the distortion energy per unit cell in covalent semiconductors is the following

$$E_{cell} = \frac{1}{2}\alpha \left(\frac{3}{4d^2}\right) \sum_{i=1}^4 \left[\Delta \left(\mathbf{r}_i^1 \cdot \mathbf{r}_i^1\right)\right]^2 + \frac{1}{2} \sum_{s=1}^2 \beta^s \left(\frac{3}{4d^2}\right) \sum_{i=1}^4 \sum_{j>i} \left[\Delta \left(\mathbf{r}_i^s \cdot \mathbf{r}_j^s\right)\right]^2. \quad (1)$$

Here α_{ij} and β_{ij} represent the force constants for bond length and bond angle distortions in the crystal, respectively; $\mathbf{d}$ and $\mathbf{r}$ are the equilibrium and real bond length, respectively;

$\Delta(\mathbf{r}_i \cdot \mathbf{r}_j)$ is the change of the dot product of the two bond vectors that start at the common atom, point along the bond directions, and end at the *first-neighbor* atoms.

1.2 Phonon spectrum

As a result of thermal fluctuations every i -th atom is displaced from its equilibrium position $\mathbf{d}_i$ by an amount $\mathbf{u}(\mathbf{d}_i)$. The total kinetic energy of the lattice becomes

$$T = \frac{1}{2} \sum_i \sum_{\xi=x,y,z} M_i \left(\frac{d u_\xi(\mathbf{d}_i)}{d t} \right)^2. \quad (2)$$

The total potential energy of the crystal is assumed to be some function of the instantaneous positions of all atoms. Within the *harmonic approximation* the corresponding equation of motion has the following form

$$M_i \frac{d^2 u_\xi(\mathbf{d}_i)}{d t^2} = - \sum_j \sum_{\zeta=x,y,z} \frac{\partial^2 E_T}{\partial u_\xi(\mathbf{d}_i) \partial u_\zeta(\mathbf{d}_j)} u_\zeta(\mathbf{d}_j). \quad (3)$$

Here E_T is the total distortion energy given by summation of Eq. (??) over all unit cells. Assuming the harmonicity of vibrations, the solution of the equation of motion (??) may be represented as

$$\mathbf{U}(\mathbf{d}_i, \omega) = \sqrt{M_i} \mathbf{u}(\mathbf{d}_i) \exp(-i\omega t). \quad (4)$$

Thus the elements of *dynamical matrix* to be solved for eigenvalues corresponding to ω^2 have the following form

$$D_{\xi\zeta}(\mathbf{d}_i, \mathbf{d}_j) = (M_i M_j)^{-\frac{1}{2}} \frac{\partial^2 E_T}{\partial u_\xi(\mathbf{d}_i) \partial u_\zeta(\mathbf{d}_j)} \quad (5)$$

The eigenvalues of the dynamical matrix in NEMO-3D are always searched by PARPACK.

2 Input file

In order to be able to compute phonon dispersion you should specify in **Execution parameters** tree both

-*Calculate phonon dispersion*

and

-*Calculate strain distribution.*

The later option is absolutely necessary, since otherwise you are out of the minimum of the strain energy and the results of phonon calculations would will be wrong.

Then you should go to **Phonon calculation** subtree of **Execution parameters** tree and specify

-*Phonon model*=**VFFM** for **V**alence **F**ield **F**orce **M**odel, since it is the only model implemented so far.

-*Minimal q_{ph} [nm^{-1}]*

and

-*Maximal q_{ph} [nm^{-1}]*

are three-component vectors of initial and final phonon wave vector. The phonon spectrum will be computed in the equidistant points in the direction from minimal to maximal q_{ph} in the reciprocal space.

-*Number of q_{ph} points* is the number of points in which you want the phonon spectrum to be computed. For rough estimations of the dispersion 10 points should be enough. In order to achieve the better smoothness of phonon branches you will have to trade between computational time, which scales linearly, and your desire to see nice pictures.

-*Part of the phonon spectrum* should be set to **Lower** if you are interested in *acoustic* and **Higher** if you are interested in *optical* phonons.

-*Number of phonon branches* is recommended to be larger then 3 for better convergency. Also note that the number of phonon branches has a physical upper limit, which is three times larger then the number of atoms in the solution domain. However, due to the specifics of eigensolver, we would not recommend to ask for more then a half of the existing eigenstates for tiny systems. You may compute lower and upper parts of it in separate runs if you are interested in the whole phonon spectrum. For larger system the computational time increases dramatically if you are too curious and would like to have too many states at once. The recommended upper limit for the number of phonon branches is about a dozen.

-**Parpack** sub-subtree would be better left untouched...

-**Output** sub-subtree has only one implemented option so far: *Phonon dispersion*. If you do not specify it, you only will be able to see the corresponding output on the screen after all computations completed. If you DO specify it, the file `input_file_name.PhonDisp` will be rewritten after calculation of energy spectrum completed in every requested point of the reciprocal space.

3 Output file

The output information about computed phonon dispersion is saved in `input_file_name.PhonDisp` file if you have specified it in the **Output** sub-subtree of **Phonon calculation** subtree of **Execution parameters** tree.

This is an example of the output file `GeSi_pa.PhonDisp` for 8 lowest phonon branches of a three-dimensional array of pyramidal Ge quantum dots embedded into Si.

%8 branches of phonon dispersion:

% #	q_x, nm^{-1}	q_y, nm^{-1}	q_z, nm^{-1}		q, nm^{-1}		0-br	1-br	2-br	3-br	4-br	5-br	6-br	7-br
0	0	0	0	0	6.61827e-10	6.73645e-10	0.00168103	0.00173119	0.00173152	0.00177565	0.00178688			
1	0	0	0.0253684	0.0253684	0.000220557	0.000254217	0.000282209	0.00169869	0.00174823	0.00174912	0.00178307	0.0017895		
2	0	0	0.0507368	0.0507368	0.000432254	0.000491335	0.00053944	0.00174125	0.00177668	0.00178374	0.00180485	0.00180649		

```

3 0 0 0.0761053 0.0761053 0.000627543 0.000700047 0.000756606 0.00177192 0.00178773 0.00182736 0.00182759 0.00184651
4 0 0 0.101474 0.101474 0.000801149 0.000876681 0.000932241 0.00178976 0.00179998 0.00184964 0.00188272 0.00191156
5 0 0 0.126842 0.126842 0.000950565 0.00102279 0.00107355 0.00181356 0.00181919 0.00187484 0.00193757 0.00198882
6 0 0 0.152211 0.152211 0.00107613 0.00114194 0.00118913 0.00184726 0.00184734 0.00190359 0.00199477 0.002059
7 0 0 0.177579 0.177579 0.00118052 0.00123821 0.00128562 0.00188414 0.00189174 0.00193679 0.00205174 0.00208816
8 0 0 0.202947 0.202947 0.00126752 0.00131607 0.00136814 0.00192735 0.00194541 0.00197588 0.00210548 0.00211059
9 0 0 0.228316 0.228316 0.00134084 0.00138027 0.00144115 0.00197271 0.00200557 0.00202243 0.00213049 0.00215376
10 0 0 0.253684 0.253684 0.00140389 0.00143526 0.00150824 0.00201758 0.00206897 0.00207569 0.00214967 0.00219585
11 0 0 0.279053 0.279053 0.00145979 0.00148471 0.00157163 0.00206223 0.00213145 0.00213257 0.00216921 0.00223249
12 0 0 0.304421 0.304421 0.00151121 0.00153118 0.0016323 0.00210751 0.00218063 0.0021925 0.00219272 0.00226477
13 0 0 0.329789 0.329789 0.00156006 0.00157627 0.00169053 0.00215365 0.00220529 0.00222256 0.00223252 0.00228604
14 0 0 0.355158 0.355158 0.00160744 0.0016207 0.00174629 0.00219862 0.00220773 0.00220952 0.00223209 0.00231815
15 0 0 0.380526 0.380526 0.00165352 0.00166438 0.00179924 0.00216534 0.00216878 0.00224668 0.00225257 0.00227263
16 0 0 0.405895 0.405895 0.00169748 0.00170626 0.00184854 0.00211923 0.00212568 0.00222733 0.00227091 0.00229348
17 0 0 0.431263 0.431263 0.00173684 0.00174375 0.00189214 0.00207715 0.00208555 0.0021863 0.00228861 0.00233422
18 0 0 0.456632 0.456632 0.00177161 0.00182503 0.00204574 0.00205564 0.00215483 0.00230115 0.00236492
19 0 0 0.482 0.482 0.00177734 0.00178202 0.00193768 0.00203399 0.00204451 0.00214261 0.00230563 0.00237651

```

The first column counts points in the reciprocal space; phonon wave vector x, y, z components in nm^{-1} are written in the next 3 columns; the 5-th column contains the amplitude of the phonon wave vector in nm^{-1} ; and the energy of phonon modes in eV is written in the rest 8 columns.

The format of the ASCII `input_file_name.PhonDisp` file is designed to allow to plot phonon dispersion curves in matlab. In order to plot the first two dispersion curves from the output file `GeSi_pa.PhonDisp` with blue dots connected with lines you should type the following commands in matlab:

```

>> e=load('GeSi_pa.PhonDisp');
>> figure
>> plot(e(:,5),e(:,6),'b.-');
>> hold on
>> plot(e(:,5),e(:,7),'b.-');

```

This output file is rewritten after calculation of the energy spectrum is completed in every requested point of the reciprocal space. Thus if you see that some rows of the file are filled with zeros, it means that the calculations have not been completed for this point yet.

4 Examples

4.1 Dispersion of bulk materials

The bulk material may be built from translation in all 3 directions of one unit cell. That is you should specify only one shape with $a_0 \leq L_x = L_y = L_z < 2a_0$, where a_0 is unstrained lattice constant of the material.

`Si.xml` is the example of the input file to compute 12 lower branches of the phonon dispersion in [001] crystallographic direction of bulk Si. You may change the *Part of the phonon spectrum* to *Higher* and together with the *Lower* one it would complete the whole phonon spectrum of bulk Si.

Note that since the actual unit cell consists of 4 elementary cells, i.e., it contains 8 atoms instead of 2, the computed spectrum corresponds to the twice folded real one (see

the output file `Si.PhonDisp`) and there are 24 eigenstates instead of 6 phonon branches. You should “unfold” it manually.

4.2 Phonon dispersion in nanostructures

The calculation of phonon spectrum may be done for all types of nanostructures generally implemented in NEMO-3D code: quantum wells, nanowires, quantum dots and their regimented 1D, 2D, and 3D arrays. Also, you may specify more complicated structures using combination of different shapes.

`GeSi_pa.xml` is the example of the input file to compute 12 lower branches of the phonon dispersion in [001] crystallographic direction of a three-dimensionally regimented array of pyramidal Ge quantum dots in Si with atomically smooth interfaces. The base of the dot $L_x = L_y = 6.518\text{nm}$ (i.e., 24 monolayers) and its height $L_z = 3.26\text{nm}$ (i.e., 12 monolayers) are specified in Shape 2. The periods $D_x = D_y = 13.04\text{nm}$ (i.e., 48 monolayers), and $D_z = 6.518\text{nm}$ (i.e., 24 monolayers) are specified in Shape 1. The fragment of the corresponding output file `GeSi_pa.PhonDisp` was shown in the previous section.

5 Conclusion

The phonon part of the NEMO-3D is still under development... ;)

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